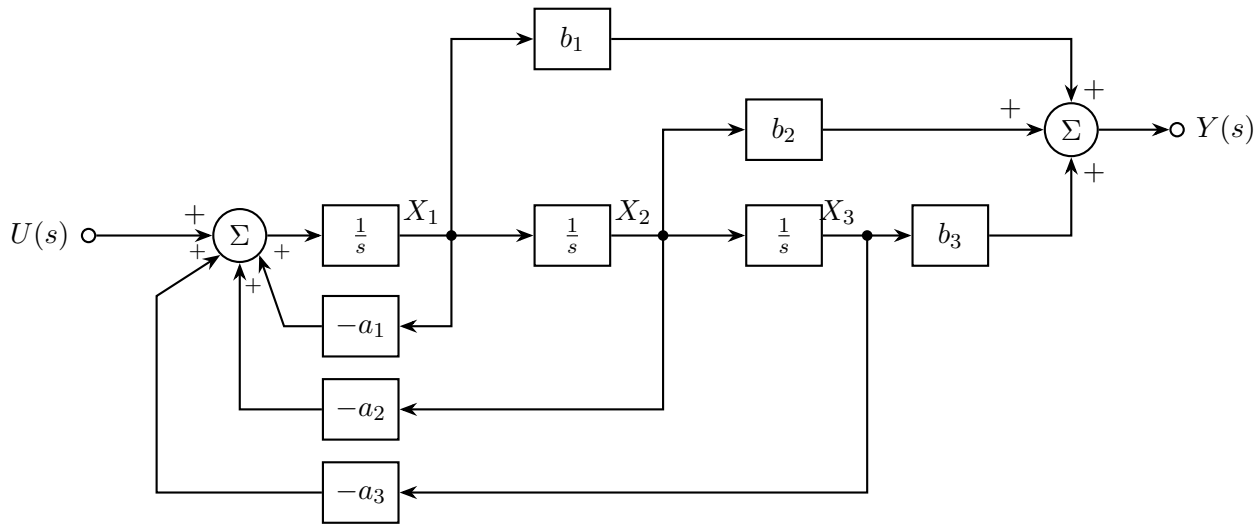


Note that the HW is 6 problems; final grade is out of 70.

Problem 1 — 10 points

For the following block diagram, called “control canonical form”, calculate the transfer function for this system. Note that a_i and b_i are constants.



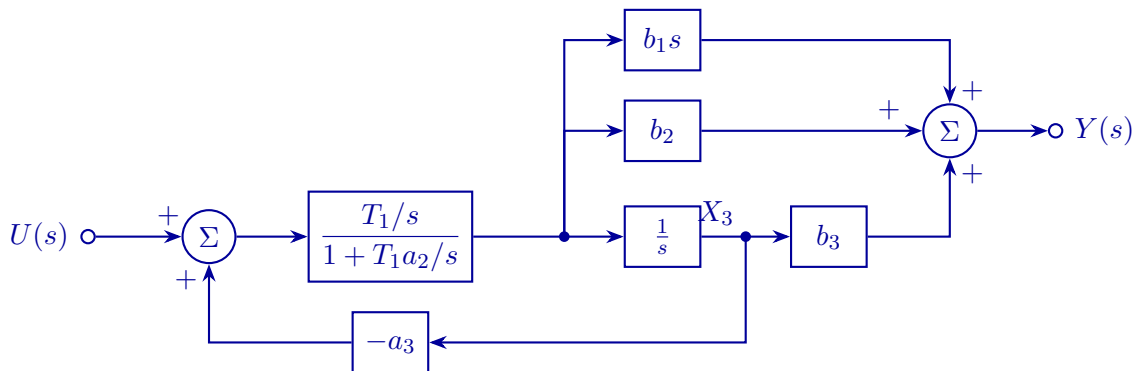
Let

$$T_1 = \frac{1/s}{1 + a_1/s}$$

and

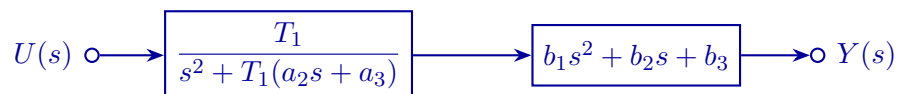
$$T_2 = \frac{T_1/s}{1 + T_1 a_2/s}$$

which reduces the system to



Now

$$T_3 = \frac{\frac{T_1/s}{s + T_1 a_2}}{1 + \frac{a_3 T_1/s}{s + T_1 a_2}} = \frac{T_1}{s^2 + T_1(a_2 s + a_3)}$$

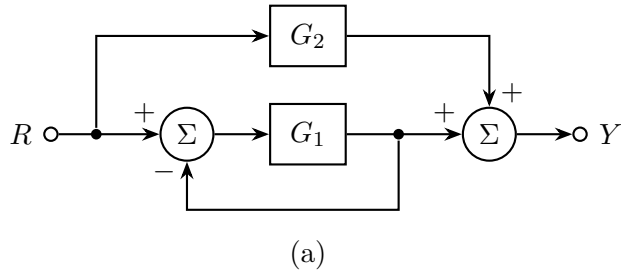


So

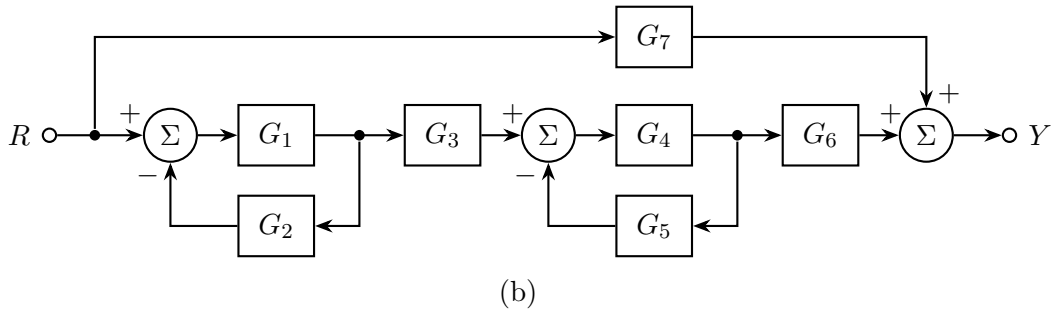
$$\begin{aligned}
 \frac{Y(s)}{U(s)} &= \frac{T_1(b_1s^2 + b_2s + b_3)}{s^2 + T_1(a_2s + a_3)} \\
 &= \frac{\frac{1}{s+a_1}(b_1s^2 + b_2s + b_3)}{s^2 + \frac{1}{s+a_1}(a_2s + a_3)} \\
 &= \boxed{\frac{b_1s^2 + b_2s + b_3}{s^3 + a_1s^2 + a_2s + a_3}}
 \end{aligned}$$

Problem 2 — 15 points

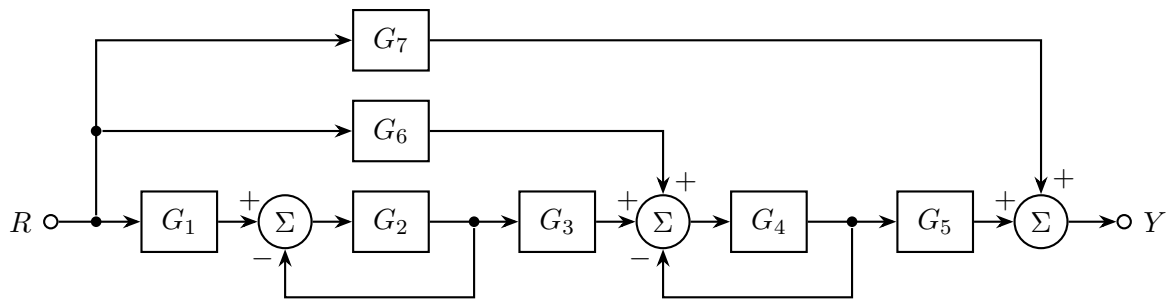
For the following block diagrams, find the transfer function Y/R .



$$\begin{aligned}
 T_1 &= \frac{G_1}{1 + G_1} \\
 \frac{Y}{R} &= T_1 + G_2 = \boxed{\frac{G_1 + G_2(1 + G_1)}{1 + G_1}}
 \end{aligned}$$



$$\begin{aligned}
 T_1 &= \frac{G_1}{1 + G_1G_2} \\
 T_2 &= \frac{G_4}{1 + G_4G_5} \\
 \frac{Y}{R} &= T_1G_3T_2G_6 + G_7 = \boxed{\frac{G_1G_3G_4G_6 + G_7(1 + G_1G_2)(1 + G_4G_5)}{(1 + G_1G_2)(1 + G_4G_5)}}
 \end{aligned}$$



(c)

$$\begin{aligned}
 T_1 &= \frac{G_1 G_2 G_3}{1 + G_2} + G_6 \\
 \frac{Y}{R} &= \frac{T_1 G_4 G_5}{1 + G_4} + G_7 \\
 &= \frac{\left(\frac{G_1 G_2 G_3}{1 + G_2} + G_6 \right) G_4 G_5}{1 + G_4} + G_7 \\
 &= \boxed{\frac{G_1 G_2 G_3 G_4 G_5 + (1 + G_2) G_4 G_5 G_6 + (1 + G_2)(1 + G_4) G_7}{(1 + G_2)(1 + G_4)}}
 \end{aligned}$$

Problem 3 — 5 points

Use MATLAB to plot the step response for the following transfer function. Calculate the natural frequency and damping ratio using time-domain data. Use the **damp** command to verify.

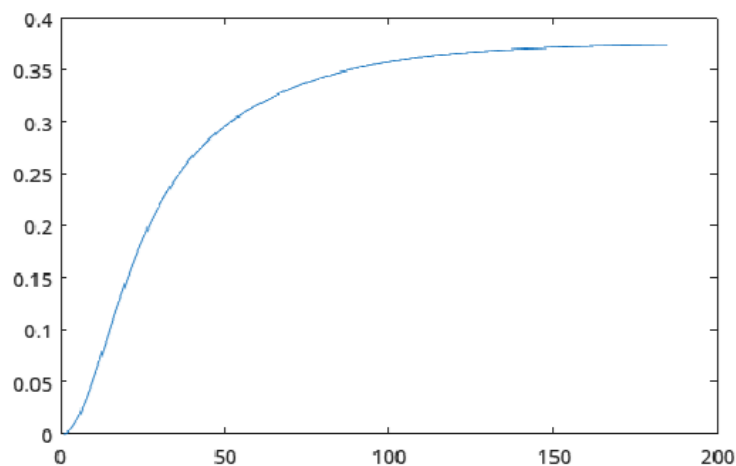
$$F(s) = \frac{2(s + 3)}{(s + 1)(s^2 + 6s + 16)}$$

```

1  s = tf('s');
2  F = 2*(s + 3)/((s+1)*(s^2 + 6*s + 16))
3
4  u = step(F);
5  plot(u);
6
7  [omega, zeta, poles] = damp(F);

```

produces



```

omega = 3×1
    1.0000
    4.0000
    4.0000

zeta = 3×1
    1.0000
    0.7500
    0.7500

poles = 3×1 complex
   -1.0000 + 0.0000i
   -3.0000 + 2.6458i
   -3.0000 - 2.6458i

```

Problem 4 — 10 points

Derive the equations for a DC servo motor and show that the final equation is

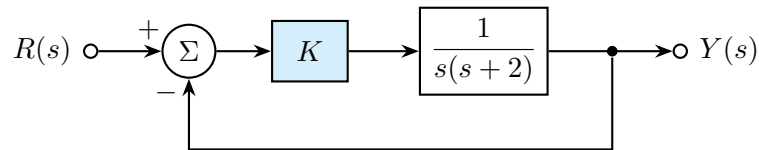
$$J_m \ddot{\theta}_m + \left(b + \frac{K_t K_e}{R_a} \right) \dot{\theta}_m = \frac{K_t}{R_a} v_a$$

Assume that $J_m = 0.01 \text{ kg} \cdot \text{m}^2$, $b = 0.001 \text{ N} \cdot \text{m} \cdot \text{s}$, $K_e = 0.02 \text{ V} \cdot \text{s}$, $K_t = 0.02 \text{ N} \cdot \text{m/A}$, $R_a = 10 \text{ } \Omega$.

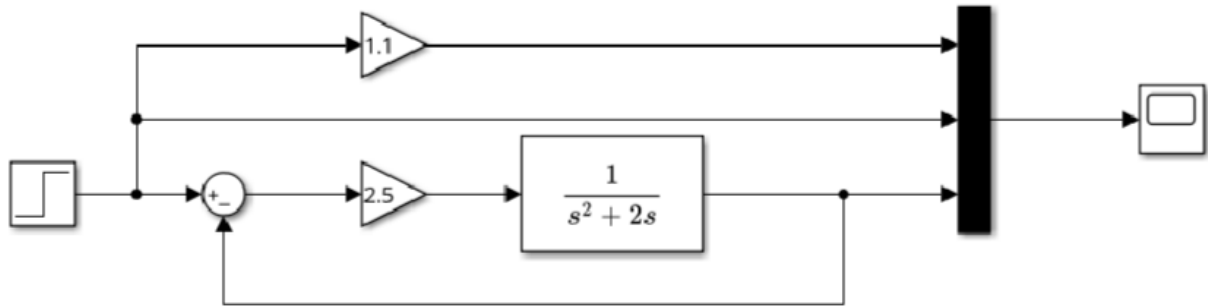
- Find the transfer function between the applied voltage and motor speed.
- What is the steady-state speed after a voltage $v_a = 10 \text{ V}$ has been applied?
- Find the transfer function between the applied voltage and shaft angle.
- Represent the above equation in state-space.
- Model the above equation in Simulink and find the steady-state velocity for applied voltages between 1 V and 10 V.

Problem 5 — 10 points

For the unity feedback system shown below, use Simulink or MATLAB to design a proportional controller with gain K that results in a maximum overshoot of 10% in response to a unit step.



Letting $k = 2.5$,

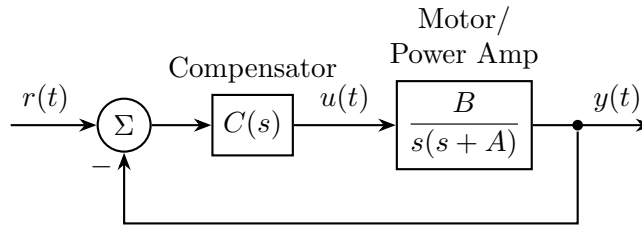


yields



Problem 6 — 20 points

The purpose of this problem is to design an analog compensator and perform MATLAB/Simulink simulation for position control of a DC motor as shown below:



where $A = 22$, $B = 120$. The inputs/outputs in the figure are defined as follows:

- $r(t)$: reference input in volts; 1 volt = 20 radians (a step input of 1 volt)
- $y(t)$: motor position sensor output in volts (1 volt = 20 radians)
- $u(t)$: voltage input to plant

The design problem entails finding the values of the coefficients of the compensator $C(s)$, in the form

$$C(s) = K \cdot \frac{s + a}{s + b},$$

by iteration, while satisfying the following constraints:

- The maximum input to the motor should not exceed ± 5 V under any condition.
- Minimum rise time and settling time of the motor and no overshoot.

Helpful procedure:

- Start with gain compensation K ; choose the gain to give a critically damped response.
- Consider a first-order compensator $C(s)$ that gives a critically damped response.
- Consider the use of an input filter $F(s) = \frac{d}{s + d}$ that $r(t)$ goes through before reaching the summing junction. Design d , a , b , and K to achieve the best settling time with no overshoot without violating the ± 5 V constraint.
- Plot the results using MATLAB. Save the data to the workspace using either a scope or the “To Workspace” block.
- Use a transfer function block in MATLAB for the compensator:

```
>> numC = K*[1 a];
>> denC = [1 b];
```

Change the values of K , a , and b from the workspace, and the Simulink diagram will change. Iterate the values of a and b .